

Project Scope: Two-Phase Golden Number Search for Random Forest Optimization

Overview

This project develops and validates a novel hyperparameter optimization algorithm for Random Forests that achieves provably optimal tree count selection with logarithmic complexity. While Random Forests are widely used in machine learning, determining the optimal number of trees ($n_{\text{estimators}}$) typically requires expensive grid search across hundreds of configurations. Our Two-Phase Search method reduces this computational burden by $7\text{-}15\times$ compared to grid search while maintaining ϵ -optimal convergence guarantees.

Problem Statement

Current approaches to Random Forest hyperparameter tuning face a fundamental trade-off: exhaustive grid search guarantees finding the optimum but requires $O(N)$ evaluations (typically 100+ model trainings), while heuristic methods (random search, Bayesian optimization) offer no optimality guarantees. For large datasets like Covertypes (581k samples), each model evaluation costs 30-60 seconds, making grid search prohibitively expensive (1-2 hours per experiment). This computational bottleneck limits iterative model development and systematic experimentation.

Proposed Solution

We introduce a Two-Phase Golden Number Search that exploits the characteristic plateau behavior of Random Forest performance curves:

Phase 1 (Exponential Bracketing): Rapidly identify the plateau region by testing $k = 10, 20, 40, 80, \dots$ until diminishing returns are detected using a window-based gradient threshold (Theorem 2).

Phase 2 (Binary Refinement): Apply binary search within the bracketed region to pinpoint the optimal tree count with precision δ .

The method achieves $O(\log N + \log(1/\delta))$ complexity with ε -optimal guarantees (Theorems 1-3), matching information-theoretic lower bounds (Theorem 7). Bidirectional convergence validation ensures robustness.

Theoretical Contributions

Theorem 1: Exponential search completes in $O(\log(k^*/k_{\min}))$ evaluations

Theorem 3: Combined two-phase approach achieves ε -optimality

Theorem 7: Proof of asymptotic optimality (matches $\Omega(\log N)$ lower bound)

Empirical Validation Goals

This study validates theoretical claims through comprehensive benchmarking:

Correctness: Verify Two-Phase Search finds optimal $k^* \geq 95\%$ of the time (within 10 trees of ground truth)

Efficiency: Demonstrate $7\text{-}15\times$ speedup over grid search across 3-5 datasets (Covertypes, MNIST, Adult)

Competitiveness: Show superiority to Random Search and Bayesian Optimization in evaluations, time, and reliability

Robustness: Confirm noise tolerance under cross-validation variance

Scalability: Validate $O(\log N)$ complexity empirically across varying search spaces

Technical Scope

We are building a modular Python benchmarking suite with:

Four search method implementations (Grid, Random, Bayesian, Two-Phase)

Automated experiment orchestration across datasets and random seeds

Statistical validation (paired t-tests, Cohen's d effect sizes)

Publication-quality result generation (5 tables, 3 figures per manuscript guidelines)

Deliverables: Reproducible codebase, empirical results tables, convergence visualizations, and manuscript-ready statistical analysis demonstrating that Two-Phase Search is both theoretically optimal and practically superior for Random Forest hyperparameter optimization.